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Personal page

Summary

Bioinformatics researcher with a keen interest in machine learning and deep learning architectures to analyze bulk, single-cell, and spatial transcriptomics through high-performance computing.

Education

MS in Bioinformatics — *Charles University*, *Sept 2024 – present*

- **Comparative analysis of non-melanoma skin cancers**: (Diploma thesis) Investigating the transcriptomic landscape of non-melanoma skin cancers using a multi-scale approach. Integrating data across platforms, from global bulk transcriptomics to high-resolution Xenium spatial transcriptomics in human skin samples.
- **Inhibiting SLC6A20 Transport Using Caver-Based Tunnel Analysis**: Team project focused on structure-based drug design. We employed Caver-based tunnel analysis to characterize transport pathways and combined it with computational tools, including AutoDock Vina, CaverDock, and P2Rank, to predict and evaluate candidate inhibitory compounds, see Repository.
- **Model of $NF-\kappa B$ Dynamics:** Simulated immune regulation via ODEs; performed sensitivity analysis and bifurcation mapping to identify pathological states, see Repository
- **Spotify Data Analysis:** Team project where we applied scRNA-seq analogy to music data. Applied TF-IDF embeddings and an interactive visualization of user sound-feature spaces.

BS in Molecular Biology and Biochemistry of Organisms — *Charles University*, *Sept 2021 – June 2024*

- **Bachelor thesis:** Algorithms for integration of single-cell transcriptomic data

  Benchmarked 7 algorithms (scVI, Seurat, Harmony, FastMNN, Scanorama, STACAS, CanSig) for scRNA-seq integration on HPC clusters. Implemented end-to-end pipelines including QC, normalization, and dimensionality reduction (PCA, tSNE, UMAP).
- **Molecular Biology and Protein Analysis**: Gained hands-on experience in recombinant protein production and purification, plasmid DNA isolation and sequencing, nucleic acid analysis (DNA/RNA isolation, RT-PCR, qPCR), fusion PCR, and reporter gene assays in bacterial and yeast systems.

Experience

Faculty of Natural Sciences of Comenius University *Bratislava, Slovakia*
April 2026

- **Workshop Instructor**: Conducted a PyMOL workshop introducing structural bioinformatics to 40 high school students, as part of DNA Day at Comenius University. Materials are available on this dedicated webpage.

Institute of Molecular Genetics of the CAS *Prague, Czechia*
2023 – present

- **Bulk RNA-seq:** Analyzed datasets in R using ComBat, limma, and clusterProfiler.
- **scRNA-seq:** Conducted heterogeneity and DEG analysis using Seurat and Scanpy.
- **Spatial Transcriptomics:** Utilizing Xenium in situ data for Diploma thesis.
- **Infrastructure:** Working on high-performance computing clusters of Czech Academy of science and MetaCentrum Virtual Organization.

Skills

Languages: Slovak - Native, Czech - C2, English - C1, Russian - C1

Programming & Tools: R (Seurat, Bioconductor, Harmony, FastMNN, ComBat), Python (Numpy, Pandas, Seaborn, Pyplot, PyTorch, Scanpy, scVI, scikit-learn, CanSig, Squidpy, SpatialData)

Analysis: Single-Cell RNA-seq, Bulk RNA-seq, Spatial Transcriptomics, ODE Modeling, Germline and Somatic variant calling

Computing: High-Performance Computing (HPC/Slurm), Bash Scripting, LaTeX, Git

Courses

Single Cell Transcriptomics — Certificate *IMG & ELIXIR Czech Republic*
Dec 2025

- Built R workflows for droplet-based scRNA-seq: from raw counts to cell annotation and functional enrichment. Applied QC, normalization, dataset integration, and DE analysis.

Charles University
Deep Learning & Neural Networks — Certificate *Oct 2024 - Jan 2025*

- Implemented FFN, RNN, CNN, and Transformer architectures in PyTorch for CV and NLP tasks. Optimized training pipelines using SGD/Adam, Dropout, and Batch Norm.

Charles University
Genomics – Approaches and Algorithms *Oct 2024 - Feb 2025*

- Developed a reproducible Bash pipeline for NGS germline variant calling (BWA, Samtools, BCFtools).

Charles University
Machine Learning in Biology *Feb 2025 - May 2025*

- Applied ML models to complex biological datasets with an emphasis on method selection and interpretation.

University of Helsinki
Elements of AI — Certificate *Oct 2024 – Jan 2025*

- Completed foundational training in AI theory, search algorithms, and practical machine learning.